

Set Difference (–) in Relational Algebra

1. What is Set Difference?

Set Difference is a fundamental set operation in Relational Algebra, represented by the mathematical symbol . It follows the core concepts of traditional set theory.

★ **DEFINITION**

A – B returns all elements/tuples that are present in **A** but **NOT** present in **B**.

$$A - B = A \text{ but NOT } B$$

2. Simple Set Theory Example

Consider two discrete sets:

Given Sets:

$$S_1 = \{1, 2, 3\}$$

$$S_2 = \{3, 4\}$$

Evaluating $S_1 - S_2$:

Take everything in S_1 and remove elements present in S_2 .

$$S_1 - S_2 = \{1, 2\}$$

- **3 is removed** because it exists in both sets.
- **4 is ignored** because it is not present in S_1 .

3. Set Difference is NOT Commutative ★★★★★

A binary operation is *commutative* if changing the order of operands does not change the result ($A - B = B - A$). Set Difference **fails** this property.

Using the previous sets:

- $S_1 - S_2 = \{1, 2\}$
- $S_2 - S_1 = \{4\}$

🏆 **Golden Rule:** $S_1 - S_2 \neq S_2 - S_1$ (Set Difference is NOT Commutative!)

Operation	Symbol	Commutative?
Union	∪	✓ Yes ($A \cup B = B \cup A$)
Intersection	∩	✓ Yes ($A \cap B = B \cap A$)
Set Difference	–	✗ No ($A - B \neq B - A$)

4. Set Difference in Relational Algebra

In Relational Algebra, $R - S$ returns tuples that are present in relation **R** but absent from relation **S**.

Example:

Relation: Student

Relation: Employee

Roll_No	Name
1	A
2	B
3	C

Roll_No	Name
1	A
7	E

Query: **Student – Employee**

- Tuple (1, A) → Present in both → **Removed**
- Tuple (2, B) → Present only in Student → **Kept**
- Tuple (3, C) → Present only in Student → **Kept**

Result: **Student – Employee**

Roll_No	Name
2	B
3	C

5. Union Compatibility Requirements ★★★★★

Before performing Set Difference ($R - S$), the two relations **MUST** be Union-Compatible.

UNION COMPATIBILITY RULES

- Condition 1 (Same Degree):** Both relations must have the exact same number of attributes/columns.
Valid: $R(A, B, C)$ and $S(X, Y, Z)$ | Invalid: $R(A, B, C)$ and $S(X, Y)$
- Condition 2 (Compatible Domains):** Corresponding attributes must have matching data types.
Valid: $R(A: \text{Integer}, B: \text{String})$ and $S(X: \text{Integer}, Y: \text{String})$
Invalid: $R(A: \text{Integer})$ and $S(X: \text{Date})$

6. Finding "Student but NOT Employee" & Reverse Query

This is a classic university and GATE exam query pattern.

Case 1: "Student but NOT Employee"

Step 1: Project Name attribute from both relations to ensure compatible schemas.

$$\pi_{\text{Name}}(\text{Student}) - \pi_{\text{Name}}(\text{Employee})$$

Evaluation:

$$\{A, B, C\} - \{A\} = \{B, C\}$$

Name
B
C

Case 2: "Employee but NOT Student"

Reverse the order: The English phrasing directly dictates the operand positions.

$$\pi_{\text{Name}}(\text{Employee}) - \pi_{\text{Name}}(\text{Student})$$

Evaluation:

$$\{A\} - \{A, B, C\} = \emptyset \text{ (Empty Set)}$$

Name
(No rows returned)

🧠 English-to-Algebra Rule: "A but NOT B" → $A - B$

8. Why Projection (π) May Be Needed Before Difference

Set difference compares **entire, complete tuples** across all attributes.

⚠ IMPORTANT STRUCTURAL CAVEAT

If you evaluate *Student* – *Employee* directly without projection, it compares complete rows like (1, 'A') vs (4, 'A'). Because the primary keys differ, (4, 'A') will **NOT** eliminate (1, 'A') even though the names match!

To compare purely by attribute values (e.g. Name), you must project the target column first using π_{Name} .

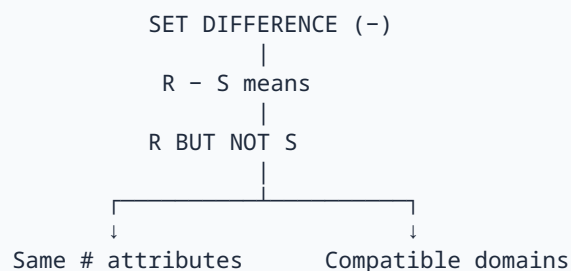
9. Operator Function Comparison

Operator	Symbol	Primary Function	SQL Equivalent
Projection	π	Selects specific Columns	SELECT DISTINCT column
Selection	σ	Filters specific Rows based on predicate	WHERE condition
Set Difference	–	Tuples in 1st relation but NOT in 2nd	EXCEPT / MINUS

⚠ Common Exam Traps & Pitfalls

WATCH OUT FOR THESE HIGH-FREQUENCY EXAM TRAPS:

- **Trap 1: Assuming Commutativity:** Writing $A - B = B - A$ will result in zero marks. $A - B \neq B - A$.
- **Trap 2: Expecting a Formula for Row Count:** Unlike Cartesian product ($m \times n$), there is no static formula for $|R - S|$. Range: $0 \leq |R - S| \leq |R|$.
- **Trap 3: Incompatible Schema:** Attempting $R(A, B) - S(A, B, C)$ directly is illegal due to union incompatibility.
- **Trap 4: Reversing Operands:** "Student but not Employee" → *Student* – *Employee* (NOT *Employee* – *Student*).



🧠 Quick Structural Revision Map:

🏆 ONE-LINE EXAM DEFINITION

Set Difference ($-$) is a binary relational algebra operation that returns all tuples present in the first relation but not in the second relation, provided the two relations are union-compatible.